When Failure Memories Help or Hurt: Conservative Selection for Tool-Using LLM Agents

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Abstract

Historical failure memories can change a tool-using language-model agent's recovery decision, but retrieval similarity does not establish that a memory is applicable to the current failure. We study PROPER, a conservative selector that uses only observable execution evidence to rank candidate memories and replaces the TF-IDF Rank-1 candidate only when an alternative is supported and decisively preferable. Across two instruction-tuned models, three failure strata, and a frozen 403-target changed-selection population, PROPER improves paired Recovery Validity over TF-IDF by 3.72 percentage points for Qwen3-8B and 5.46 points for Mistral-7B-Instruct-v0.3. It also outperforms frozen Dense and LLM-Judge baselines in both models. These relative gains do not imply that memory is universally useful: all evaluated memory methods remain below No Memory overall for Mistral, and timeout or authorization strata contain zero or weak effects. Ablations identify the conservative gate as the clearest supported component, while contradiction checking has no stable independent behavioral benefit. Descriptive risk and concentration analyses further expose negative transfer and model-specific effect concentration. **TODO-V1: blinded evaluator audit and inter-annotator agreement**. Our conclusions are limited to the evaluated benchmark, memory bank, models, and one-step recovery horizon.

1 Introduction

Tool-using language-model agents often face recoverable execution failures: an argument is missing, authorization is temporarily denied, or a tool times out. A natural response is to retrieve a successful historical recovery and insert it into the agent context. Yet a memory that is lexically similar may encode the wrong operation, assume an unavailable precondition, target an incompatible tool, or recommend a retry that public evidence already contradicts. Historical memory can therefore help, do nothing, or induce negative transfer.

This paper asks whether observable failure evidence can support a more conservative decision: treat similarity retrieval as a proposal, and replace its Rank-1 memory only when a candidate is applicable and decisively better supported. PROPER implements this decision without benchmark names, hidden fault plans, gold actions, evaluator outcomes, or prior model outcomes.

Our evidence supports a deliberately bounded conclusion. Relative to TF-IDF, Dense retrieval, and a fixed LLM applicability judge, PROPER improves one-step Recovery Validity for both evaluated models on a pre-frozen changed-target population. Absolute memory utility, however, differs sharply by model and failure type. In particular, memory helps Qwen overall but remains worse than No Memory for Mistral, and some strata show no behavioral gain despite selection changes.

We make three contributions: (1) an observable-only, fail-closed selector that separates candidate support, compatibility and preconditions, contradiction signals, and a conservative replacement gate; (2) a frozen two-model, three-stratum paired evaluation against retrieval, reranking, no-memory, oracle, and component-ablation controls; and (3) a risk-centered diagnosis of negative transfer, selection-to-behavior gaps, and effect concentration. We do not claim external or multi-step generalization.

2 Related Work

Agent memory and experience reuse. TODO-CITATIONS: add primary-source literature on episodic memory, reflection, and experience retrieval for language-model agents, distinguishing memory storage from

post-failure applicability selection.

Retrieval and reranking. TODO-CITATIONS: position TF-IDF, dense retrieval, and LLM reranking as increasingly expressive relevance estimators while emphasizing that relevance and executable applicability are different targets.

Failure recovery and selective decision making. TODO-CITATIONS: connect tool-execution feedback, abstention/selective prediction, and conservative policy improvement to the observable-evidence gate studied here.

Our scope is narrower than general long-term agent memory or multi-step planning: we select one historical memory after one observable tool failure and evaluate the next recovery decision.

3 Problem Formulation

Let o denote the observable post-failure state: the user instruction, public tool schemas, failed action and arguments, public error or return, retry budget, and observable history. A source-isolated retriever returns ordered memory candidates $C = (c_1, \ldots, c_k)$. A selector returns one of three actions: keep c_1, select an alternative c_j, or abstain. Abstention preserves c_1 as the retrieval fallback; it does not claim that c_1 is applicable.

The selector is forbidden from reading benchmark or scenario names, hidden failure plans, recoverability/applicability labels, gold actions, evaluator outcomes, method identities, or prior model outcomes. The primary endpoint is paired binary Recovery Validity after one recovery decision. The primary population is frozen before agent generation as the union of targets where at least one non-oracle selector differs from TF-IDF Rank-1.

4 Method

PROPER represents each retrieved memory as a policy card containing its trigger evidence, required preconditions, recovery operation, proposed tool/action, repair targets, continuation policy, stop conditions, source tool, and extraction confidence. Candidate evaluation uses only o and this public card.

4.1 Evidence and compatibility

Operation-specific support prevents generic lexical overlap from being treated as applicability. Argument repair requires both trigger overlap and repair-target overlap; stopping requires matching observable stop evidence; other known recovery operations require trigger overlap. Preconditions, repair-target agreement, trigger agreement, and tool compatibility are evaluated separately.

4.2 Contradictions and conservative replacement

The selector records active contradictions such as an exhausted public retry budget or an unsafe identical retry. Candidates are ordered by fewer active contradictions, stronger evidence support, precondition status, repair-target agreement, trigger agreement, tool compatibility, extraction confidence, original rank, and a deterministic identity tie-break.

Full PROPER replaces Rank-1 only when the best alternative has no active contradiction, has positive operation-specific support, and is decisively superior to Rank-1 on a predeclared criterion. The no-gate ablation retains contradiction checking but chooses the best contradiction-free candidate. The no-contradiction ablation retains the conservative gate but removes contradictions from eligibility and ordering. No other component changes.

5 Experimental Setup

5.1 Data and conditions

The frozen evaluation contains 541 valid targets: 189 argument-omission, 175 transient-authorization, and 177 timeout cases, with a source-isolated bank of 100 memories. Conditions are No Memory, TF-IDF Rank-1, BGE-large-en-v1.5 Dense Rank-1, a fixed Qwen3-8B applicability judge over anonymous TF-IDF Top-10 candidates, PROPER, two ablations, and a descriptive applicability Oracle. The non-oracle changed-target union contains 403 targets (152/146/105 by stratum) and is primary; all 541 targets form a sensitivity population.

5.2 Models and execution

We evaluate Qwen3-8B and Mistral-7B-Instruct-v0.3 with deterministic decoding, thinking disabled, a 256-token cap, a frozen strict parser, and one post-failure decision. Identical raw prompts are cached by full identity. Historical Qwen model outputs are reused only under exact prompt, prefix, model-manifest, runtime, and decoding identity; environment outcomes are recomputed independently for every logical row. Invalid outputs remain in the intention-to-treat evaluation under the frozen fallback.

5.3 Statistics

For each paired comparison, we report positive and negative discordant counts, the treatment-minus-baseline risk difference, a 10,000-replicate paired-instance percentile-bootstrap 95% interval, and an exact two-sided McNemar test. Holm correction is applied within pre-frozen primary, strong-baseline, and ablation families. Oracle, risk diagnostics, failure-stratum breakdowns, and concentration analyses are descriptive unless explicitly identified otherwise.

5.4 Evaluator reliability

The automated environment evaluator is frozen. A method-blinded packet contains all 280 unique discordant formal pairs plus a stratified 20% sample of 4,026 non-discordant pairs, totaling 1,099 review pairs. Both reviewer templates are currently unlabeled. **TODO-V1: blinded evaluator audit and inter-annotator agreement**.

6 Main Results

Table 1: Recovery Validity on the frozen changed-target union ($n = 403$ per model). Counts are successes; percentages are in parentheses. Oracle is descriptive.

Model	Condition	Success	Rate (%)
Qwen3-8B	no_memory	268/403	66.5
Qwen3-8B	tfidf	351/403	87.1
Qwen3-8B	dense	335/403	83.1
Qwen3-8B	llm_judge	326/403	80.9
Qwen3-8B	proper	366/403	90.8
Qwen3-8B	proper_no_gate	327/403	81.1
Qwen3-8B	proper_no_contradiction	371/403	92.1
Qwen3-8B	oracle	376/403	93.3
Mistral-7B-Instruct-v0.3	no_memory	195/403	48.4
Mistral-7B-Instruct-v0.3	tfidf	158/403	39.2
Mistral-7B-Instruct-v0.3	dense	166/403	41.2
Mistral-7B-Instruct-v0.3	llm_judge	160/403	39.7
Mistral-7B-Instruct-v0.3	proper	180/403	44.7
Mistral-7B-Instruct-v0.3	proper_no_gate	170/403	42.2
Mistral-7B-Instruct-v0.3	proper_no_contradiction	176/403	43.7
Mistral-7B-Instruct-v0.3	oracle	173/403	42.9

Table 2: Paired PROPER–TF-IDF effects. The all-strata rows are the pre-frozen primary family; failure-stratum rows are exploratory. RD is the paired risk difference in percentage points.

Model	Stratum	n	TF-IDF	PROPER	+/−	RD	95% CI
Mistral-7B-Instruct-v0.3	all_strata	403	158	180	30/8	5.5	[2.5, 8.4]
Mistral-7B-Instruct-v0.3	argument_omission	152	95	114	25/6	12.5	[5.9, 19.7]
Mistral-7B-Instruct-v0.3	transient_authorization	146	0	0	0/0	0.0	[0.0, 0.0]
Mistral-7B-Instruct-v0.3	timeout	105	63	66	5/2	2.9	[-1.9, 7.6]
Qwen3-8B	all_strata	403	351	366	15/0	3.7	[2.0, 5.7]
Qwen3-8B	argument_omission	152	136	146	10/0	6.6	[3.3, 10.5]
Qwen3-8B	transient_authorization	146	117	122	5/0	3.4	[0.7, 6.8]
Qwen3-8B	timeout	105	98	98	0/0	0.0	[0.0, 0.0]

On the changed-target union, PROPER improves paired Recovery Validity over TF-IDF for Qwen by 3.72 percentage points (15 positive and no negative discordances; 95% CI [1.99, 5.71]) and for Mistral by 5.46 points (30/8; [2.48, 8.44]). Both are significant after correction in the pre-frozen primary family. PROPER also exceeds Dense and the LLM Judge for each model after correction.

Relative selector performance must be separated from absolute memory utility. Qwen improves from 268 No-Memory successes to 366 with PROPER. Mistral instead declines from 195 to 180: PROPER is the strongest non-oracle memory selector, but memory does not beat omitting memory for this model overall.

Failure-stratum results are heterogeneous. Argument omission provides the largest gains for both models. Qwen timeout is an exact zero (0/0 discordance), Mistral transient authorization has zero recoveries under every memory condition, and Mistral timeout has a weak interval crossing zero. We therefore do not claim uniform cross-failure effectiveness.

7 Risk and Mechanism Analysis

Table 3: Core ablations on the changed-target union. Positive RD favors full PROPER. Holm correction is within the pre-frozen ablation family.

Model	Removed component	+/−	RD (pp)	Holm p
Mistral-7B-Instruct-v0.3	contradiction	5/1	1.0	0.219
Mistral-7B-Instruct-v0.3	gate	14/4	2.5	0.0927
Qwen3-8B	contradiction	0/5	-1.2	0.125
Qwen3-8B	gate	42/3	9.7	3.46e-09

Table 4: Descriptive transfer relative to No Memory on the changed-target union.

Model	Method	+/−	RD (pp)	Repeat regress.	Parse regress.
Qwen3-8B	tfidf	93/10	20.6	0	0
Qwen3-8B	proper	105/7	24.3	0	0
Mistral-7B-Instruct-v0.3	tfidf	24/61	-9.2	58	11
Mistral-7B-Instruct-v0.3	proper	34/49	-3.7	53	23

The conservative gate is the clearest supported mechanism. Full PROPER exceeds no-gate by 9.68 points for Qwen (42/3, Holm-adjusted $p = 3.46 \times 10^{-9}$). Mistral has the same direction (+2.48 points, 14/4) but is inconclusive after family correction ($p = 0.0927$). On the 95 targets where the gate changes memory identity, descriptive effects are +41.05 and +10.53 points for Qwen and Mistral, respectively.

Contradiction checking does not show stable independent behavioral benefit: removing it yields five additional Qwen successes but four fewer Mistral successes relative to full PROPER, and neither corrected comparison is significant. We retain contradiction checking as an explicit conservative rule and audit signal, not as a separately validated innovation.

Negative transfer is model-dependent. Relative to No Memory, PROPER produces 105 improvements and seven degradations for Qwen but 34 improvements and 49 degradations for Mistral. No safety violation is observed in the frozen logical rows; this absence does not establish deployment safety. Mistral also exhibits substantially more repeated-invalid and parse regressions under memory conditions.

8 Concentration and Robustness

Table 5: Descriptive concentration and leave-one-group-out sensitivity. Min RD is the smallest paired effect after excluding one group; no selector is retrained.

Model	Grouping	Groups	Largest + share	Min RD (pp)
Qwen3-8B	failed_tool	9	66.7	1.4
Qwen3-8B	task_domain	4	73.3	1.4
Qwen3-8B	proper_selected_memory	63	66.7	1.3
Mistral-7B-Instruct-v0.3	failed_tool	9	36.7	3.1
Mistral-7B-Instruct-v0.3	task_domain	4	36.7	3.2
Mistral-7B-Instruct-v0.3	proper_selected_memory	63	26.7	3.6

Qwen's primary positive transfers are concentrated: 'get_doc' and selected memory M058 each account for 10 of 15 positive discordances. Removing either group leaves a smaller but positive paired effect (+1.43 and +1.32 points). Mistral's largest tool and memory groups account for 36.7% and 26.7% of positive transfers; their exclusions leave +3.14 and +3.59 points. Across both models, removing any single failed tool, task domain, or selected memory causes no sign reversal. These are exclusion sensitivities under frozen selections, not retraining or out-of-distribution tests.

9 Limitations and Threats to Validity

The evidence covers one benchmark, one 100-memory bank, two instruction-tuned 7B/8B models, three constructed failure strata, and one recovery decision. The primary changed-target population measures selector capacity efficiently but is not a random deployment population; the all-valid analysis is a sensitivity check. Results do not establish multi-step or external-benchmark generalization.

Automated Recovery Validity has not yet completed its human reliability audit. **TODO-V1: blinded evaluator audit and inter-annotator agreement**. Until then, evaluator reliability is an explicit submission blocker rather than an inferred property.

The Mistral result shows that a selector can improve relative to other memory methods while memory remains worse than No Memory. The contradiction ablation is inconsistent across models. No observed safety events imply neither a tight risk bound nor deployment safety. Finally, subgroup, intervention, and concentration analyses are descriptive and cannot establish causal mechanisms.

10 Conclusion

Failure-memory retrieval should be treated as a proposal whose applicability depends on observable execution evidence. In the evaluated setting, conservative selection improves recovery relative to TF-IDF, Dense retrieval, and a fixed LLM Judge for both models. The broader lesson is conditional: history helps Qwen, hurts Mistral overall, and produces zero or weak effects in some failure strata. A credible failure-memory evaluation must therefore include No Memory, negative transfer, component ablations, zero cells, and concentration diagnostics alongside average gains.